CS201 Midterm 1 Part B

20 March 2024

Olcay Taner YILDIZ

***Abstract*— Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.**

I. Question (LinkedList)

Write the method

```
LinkedList intersec(LinkedList list1,
                    LinkedList list2)
```

to find the intersection of the elements in two sorted linked lists and return a new linked list. Implement the following algorithm:

1. At the beginning of the algorithm, let say we have two nodes p1 and p2, showing the head nodes of the first and second lists respectively.
2. Compare the contents of the nodes p1 and p2;
   - If p1.data $<$ p2.data, advance p1 pointer to show next node in its list.
   - If p1.data $>$ p2.data, advance p2 pointer to show next node in its list.
   - If p1.data $=$ p2.data, put a new node with content of p1 and advance both pointers p1 and p2 in their respective lists.
3. Continue with step 2 until one of the p1 or p2 is null.

You are not allowed to use any linked list methods. You are only allowed to use attributes, constructors, getters and setters.
Contents of the first list
1 3 5 7 11 12
Contents of the second list
1 2 6 7 9 11
Contents of the results
1 7 11

II. Question (DoublyLinkedList)

Write the method

```
boolean isPalindrom()
```

which returns true if the doubly linked list is palindrom. Implement the following algorithm:

- At the beginning of the algorithm, we have two pointers p1 and p2, which shows the beginning and the end of the list respectively.
- Compare the contents of the pointers, if they are different, return false, otherwise advance the pointers p1 to next, p2 to previous.
- The algorithm finishes either p1 = p2 or p1.next = p2, in which case the method returns true.

You are not allowed to use any doubly linked list methods. You are allowed to use attributes, constructors, getters and setters.

III. Question (Stack)

Write a method which removes only the even indexed (2, 4, . . .) elements from the stack. The first element at the bottom has index 1. You are only allowed to use pop, push, isEmpty functions (Hint: Use external stack). You are not allowed to use any stack attributes such as N, top, array etc.

```
void removeEvenIndexed()
```

IV. Question (Stack)

Write the method

```
LinkedList popBottomK(int k)
```

which pops k elements from the bottom and return them as a singly linked list. You are not allowed to use any stack methods and any external stacks. You are allowed to use attributes, constructors, getters and setters. Write the method for linked list implementation.